

Fixed Geometry, Not Learned Residuals: Exposing Ego-Motion Shortcuts in Sensor-Free Pedestrian Trajectory and Intent Prediction

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Abstract—Egocentric pedestrian tracks confound pedestrian motion with camera motion. Zhang *et al.* [1] decompose the mixture using ego-speed as input and soft supervision; LIM [2] deletes speed to avoid causal confusion. We ask whether classical offline geometry can recover camera motion without any speed sensor, and whether a learned module can reproduce that geometry from tracks alone. We introduce a **standing-pedestrian variance gate** and show that three learned multi-agent residual variants all fail it, while full-affine ORB compensation evaluated at each agent’s image position passes (93.1% on PIE, 96.9% on JAAD). Fixed ORB is trajectory-neutral on PIE (ADE 60.5 vs. 60.8 px) and trajectory-improving on JAAD (62.7 vs. 83.1 px, -25%). Intent AUC rises monotonically with ego-motion information ($0.80 < 0.89 < 0.93$), suggesting a substantial share of reported intent accuracy reflects an ego-motion shortcut.

Index Terms—pedestrian prediction, ego-motion compensation, ORB, causal confusion, JAAD, PIE

I. INTRODUCTION

A camera on a moving vehicle makes every tracked object’s image-plane trajectory a mixture of pedestrian motion and camera motion: a standing pedestrian still drifts across the frame as the vehicle advances. Zhang *et al.* [1] decompose this mixture using ego-speed as input and soft supervisor; LIM [2] argues ego-speed is causally entangled with driver reaction and removes it entirely. Both require ego-vehicle state unavailable on JAAD [3], which ships no continuous odometry.

We study two candidates: (i) a *learned* multi-agent residual (LEMC) regressed from displacement statistics, and (ii) *fixed* offline ORB [4] + RANSAC [5] partial affines with pedestrian boxes masked, evaluated at each agent’s centre rather than as a global translation. To adjudicate without relying solely on downstream ADE, we introduce a **standing-pedestrian variance gate**: for stationary pedestrians while the vehicle moves, valid compensation requires $\text{Var}(\text{comp}) < \text{Var}(\text{raw})$ over the observation window.

Contributions. (1) Position-dependent full-affine ORB with a diagnostic showing why global translation under-performs ($\sim 21\%$ vs. $\sim 89\%$ gate pass, $n=300$). (2) A variance gate exposing failure of three learned variants despite superficial OBD correlation ($R^2=0.30$). (3) Fixed ORB trajectory-neutral on PIE and -25% ADE on JAAD with zero sensor dependence. (4) Stratified evidence that GT speed harms ADE during acceleration/deceleration (LIM-aligned). (5) Monotonic intent-AUC ordering: compensated $<$ raw $<$ GT speed.

II. RELATED WORK

Sensor-dependent decomposition. Zhang *et al.* [1] weight trajectory loss by ego speed and report ADE 17.41 on PIE with I3D features—not comparable to our bbox-only GRU. **Sensor deletion.** LIM [2] removes speed to break causal shortcuts between driver reaction and crossing labels. **Classical geometry.** ORB-based stabilization is standard in VO [4] but has not been benchmarked as a position-dependent compensator for bbox trajectory/intent prediction under a physical validity test. **Datasets.** PIE [6] provides OBD speed; JAAD [3], [7] provides no continuous ego-motion signal.

III. METHOD

Given observation window τ and horizon η , we predict future bbox-centre trajectories and crossing intent from track coordinates only (no raw pixels at inference).

A. Fixed ORB-affine compensation

For each consecutive frame pair we match ORB keypoints on background (pedestrian boxes masked), fit partial affine M via `estimateAffinePartial2D` under RANSAC, and evaluate at ego centre (c_x, c_y) :

$$\Delta = M \begin{bmatrix} c_x \\ c_y \\ 1 \end{bmatrix} - \begin{bmatrix} c_x \\ c_y \end{bmatrix}. \quad (1)$$

Cumulative Δ compensates observation and future tracks into a stabilized frame; the GRU predicts therein; metrics are computed in ego view after adding predicted camera shift back. ORB extraction is offline and sensor-free on both PIE and JAAD.

B. Learned LEMC (ablation)

Per-transition statistics (median, trimmed mean, IQR) across *other* agents feed a GRU producing a 2D residual, cumulatively summed and subtracted from the ego track. We test unsupervised training, OBD-magnitude auxiliary loss, and direct ORB-regression supervision.

C. Variance gate

For standing pedestrians with vehicle motion, we require $\text{Var}(\text{comp}) < \text{Var}(\text{raw})$ on observed centres and report the fraction of windows passing, overall and on the standing+moving subset.

Fixed ORB-affine compensation (sensor-free at inference)

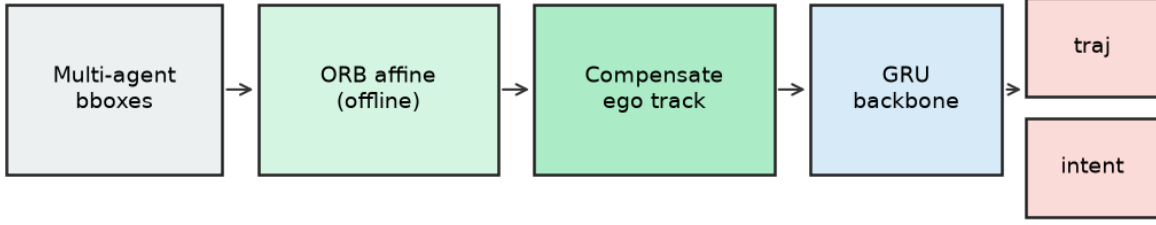


Fig. 1: Fixed ORB-affine pipeline: offline geometry compensates the ego track before GRU prediction (sensor-free at inference).

TABLE I: PIE test (mean $\pm$ std, 3 seeds).

Method	ADE	FDE	AUC
Baseline	60.81 $\pm$ 3.31	121.75 $\pm$ 9.87	0.89 $\pm$ 0.01
GT OBD speed	66.96 $\pm$ 1.28	136.89 $\pm$ 3.02	0.93$\pm$0.01
LEMC, no aux.	81.68 $\pm$ 5.13	159.91 $\pm$ 2.39	0.77 $\pm$ 0.05
LEMC + OBD aux.	85.77 $\pm$ 11.19	164.25 $\pm$ 11.33	0.80 $\pm$ 0.01
LEMC + ORB-reg.	81.73 $\pm$ 4.61	163.54 $\pm$ 7.55	0.81 $\pm$ 0.01
Fixed ORB	60.49$\pm$1.83	130.28 $\pm$ 1.31	0.80 $\pm$ 0.01

TABLE II: Standing-pedestrian variance gate (% windows passing).

Method	Dataset	All	Stand.+mov.
LEMC + OBD aux.	PIE	0.7	0.8
Fixed ORB	PIE	76.7	93.1
Fixed ORB	JAAD	71.9	96.9

IV. EXPERIMENTAL SETUP

We follow Zhang *et al.* [1]: $\tau=16$, $\eta=45$, TTE $\in [30, 60]$, 50% overlap. **PIE** [6]: 1,773 test windows, augmented track store (median 8 agents/frame). **JAAD** [3]: 295 test windows; ORB from 124 scenes with imagery; no GT-speed condition. **Metrics:** ADE/FDE, ARB/FRB@30f [1], intent AUC/F1. **Backbone:** linear embed $\rightarrow$ 2-layer GRU $\rightarrow$ traj + intent heads; three seeds, mean $\pm$ std.

V. RESULTS

A. PIE main grid

Table I summarizes PIE. Fixed ORB matches baseline ADE; every learned variant is worse, including ORB-regression supervision (81.7 vs. 60.5 px). GT speed improves intent (AUC 0.93) but degrades trajectory ADE (67.0 px).

B. Variance gate

Table II: OBD-supervised LEMC flattens 0.8% of standing+moving windows despite $R^2=0.30$ vs. OBD; fixed ORB passes at 93.1% (PIE) and 96.9% (JAAD).

TABLE III: ADE/FDE (px) by ego state, PIE seed 0.

State	n	Baseline	GT speed	Fixed ORB
Accelerating	251	58.5/127.8	88.6/173.2	52.2/122.6
Constant	1364	64.5/130.6	62.2/120.3	59.4/130.1
Decelerating	158	65.0/139.8	82.0/174.7	67.5/151.5

TABLE IV: JAAD test ($n=295$, 3 seeds).

Method	ADE	FDE	ARB	FRB	AUC
Baseline	83.10 $\pm$ 2.00	156.25 $\pm$ 4.31	44.76 $\pm$ 2.15	71.23 $\pm$ 2.08	0.47
Fixed ORB	62.68$\pm$2.34	120.11$\pm$2.82	34.80$\pm$2.25	54.17$\pm$0.91	0.44

C. ORB validation and stratified PIE

ORB translation magnitude vs. OBD speed yields $R^2=0.41$ ($n=75,268$); ORB never consumes speed (Fig. 4). During acceleration, GT speed ADE is worst (88.6 vs. baseline 58.5); fixed ORB is best (52.2)—Table III, Fig. 5.

D. JAAD cross-dataset

Table IV: with no OBD, fixed ORB reduces ADE 25% (83.1 $\rightarrow$ 62.7 px) and FDE 23%. Intent AUC is at chance (~ 0.44 – 0.47); contribution on JAAD is trajectory-only.

E. Qualitative and intent shortcut

Fig. 7 shows a standing pedestrian whose raw track drifts while ORB-compensated track stays near-stationary, and a PIE trajectory overlay on a real frame. Reading Table I intent columns: AUC rises as ego-motion information increases—fixed ORB (0.80) < baseline (0.89) < GT speed (0.93); F1 follows ($0.60 < 0.74 < 0.85$).

VI. DISCUSSION

A shared 2D shift cannot represent rotation/scale whose apparent effect depends on image position; multi-agent statistics average over a non-shared target. ORB-regression supervision does not close the gap (Table I), indicating architectural mismatch. **Limitations:** bbox-only GRU; ORB depth confound; hold-last missing affines; JAAD intent at chance; shortcut

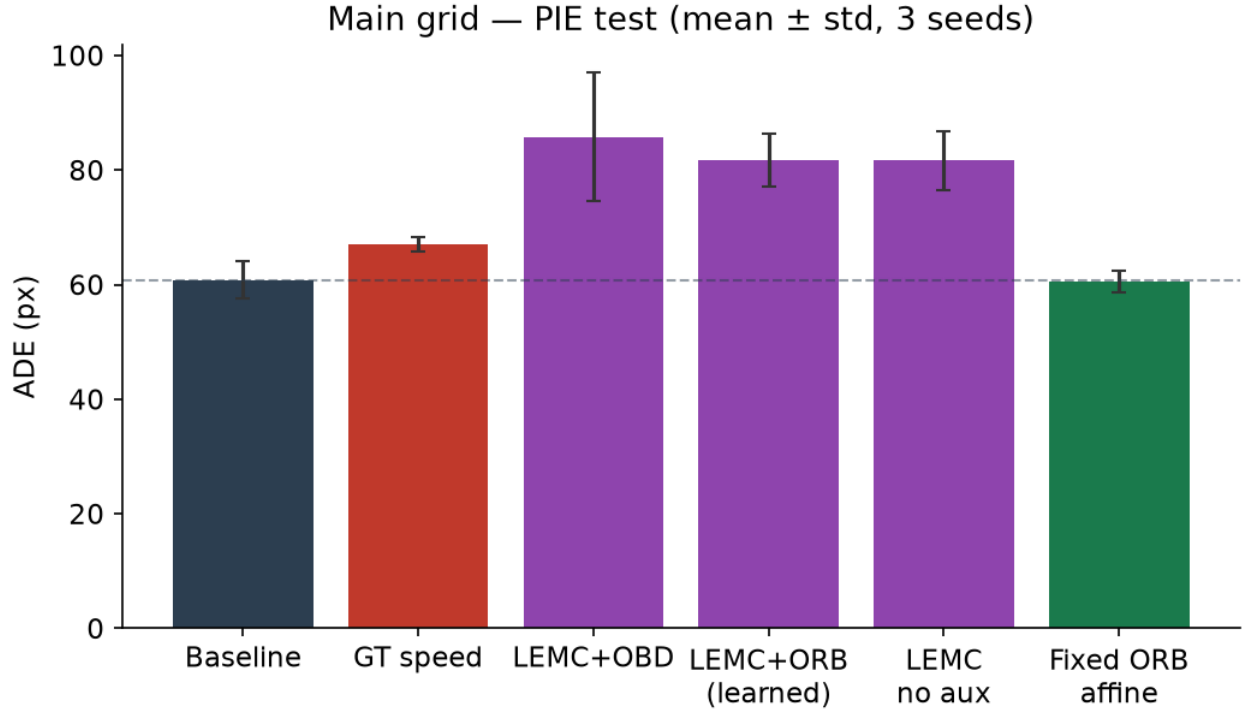


Fig. 2: PIE ADE (3 seeds). Fixed ORB ties baseline; learned LEMC variants are worse.

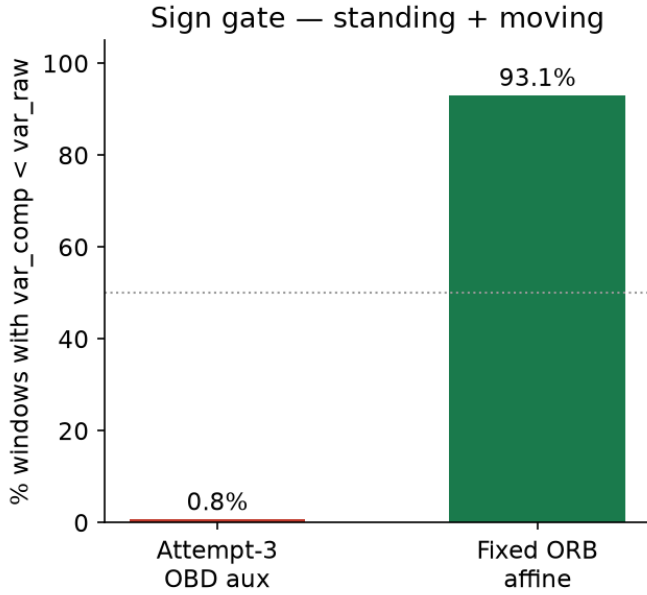


Fig. 3: Sign gate on PIE (standing+moving): learned vs. fixed ORB.

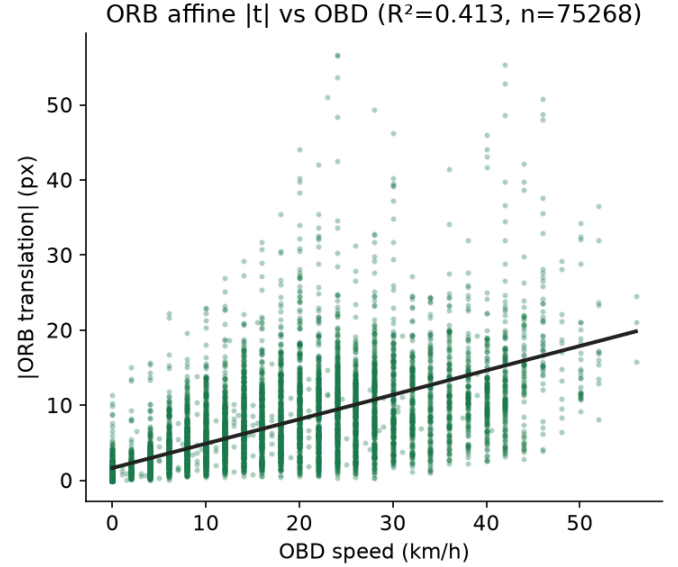


Fig. 4: ORB $|t|$ vs. OBD speed ($R^2=0.41$). Independent geometric sanity check.

finding not isolated by dedicated ablation. We do not claim SOTA vs. I3D baselines [1].

VII. CONCLUSION

Offline position-dependent ORB affine passes a physical variance test three learned alternatives fail. Trajectory-neutral on PIE, -25% ADE on JAAD, zero sensors. Intent AUC tracks

available ego-motion—sensor-free geometry helps trajectory; intent remains open.

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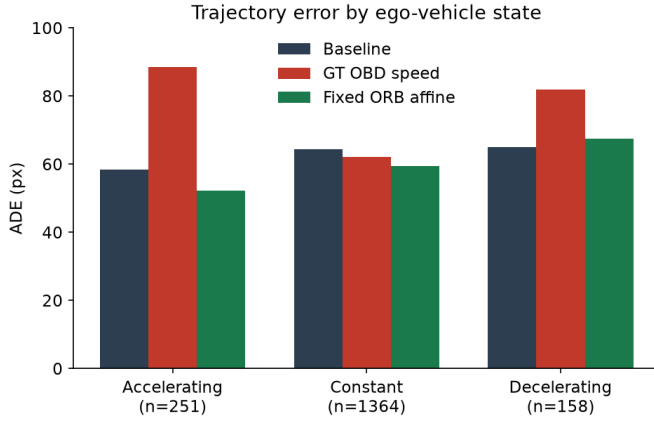


Fig. 5: ADE by ego-vehicle state (PIE, seed 0).

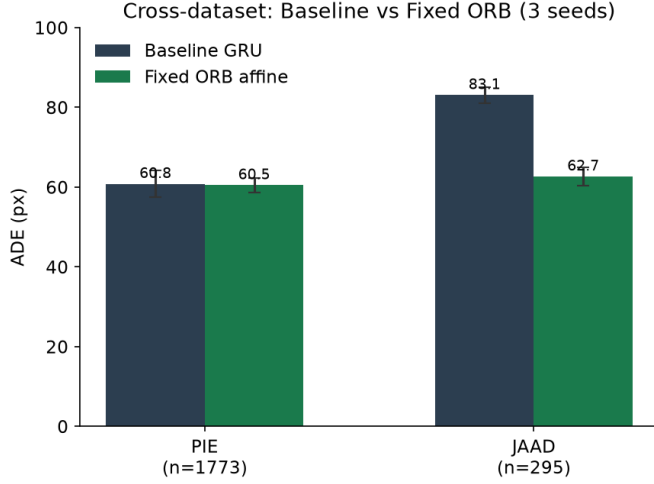
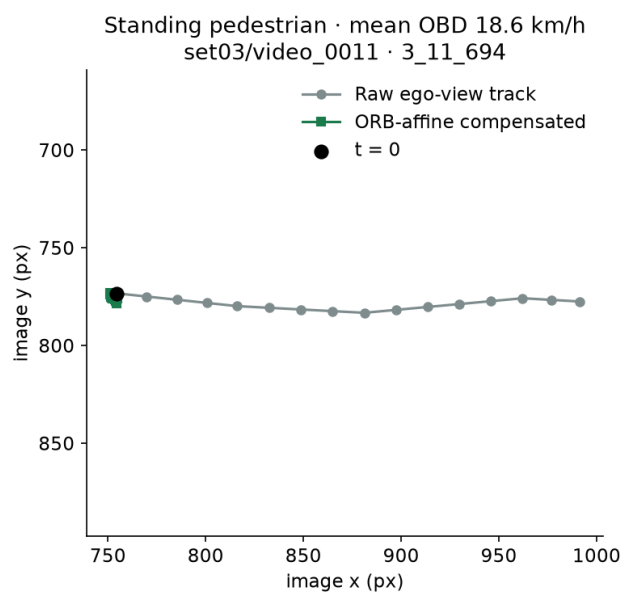


Fig. 6: Cross-dataset ADE: PIE neutral, JAAD -25% .

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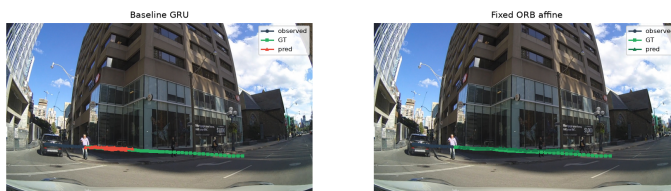


Fig. 7: Left: standing pedestrian (18.6 km/h). Right: PIE trajectory overlay on a real frame (observed, GT, predicted).